(c) Is grown in glasshouses keeping it clean of environmental pollutants.
(d) Is grown without the use of artificial fertilizers and pesticides.

U.P.P.S.C. (R.I.) 2014

Ans. (d)

Organic food is produced by farmers who emphasize the use of renewable resources and the conservation of soil and water to enhance environmental quality for future generations. The organic food is produced without using most conventional pesticides; fertilizers made with synthetic ingredients or sewage sludge, bioengineering or ionizing radiation.

Detergent and Soap

Notes

Detergent :

- Detergent is a water soluble cleansing organic compound which combines with impurities and dirt to make them soluble and differs from soap as it does not forms a scum with the salts in hard water.
- Detergent helps to remove dirt and grease from porous surfaces such as fabrics, clothes, non-treated wood and/ or non-porous surfaces such as metals, plastics and treated wood.
- Soap is used only for soft water but detergent can be used for soft water as well as hard water, because the calcium and magnesium salts of detergents are soluble in water.
- Chemically detergent is the salt of strong base and strong acid with higher molecular weight, whose anion or cation has a long chain of carbon atoms from 12 to 18.
- For example – **Sodium Lauryl Sulphate** is an anionic surfactant with a long chain of 12 carbon atoms.
- The aqueous solution of detergent is neutral so it is used to clean the clothes made from delicate fibres.
- The solution of soap is alkaline due to hydrolysis, so it is harmful for cleaning delicate clothes.

Soap :

- Soaps are salts of fatty acids whereas fatty acids are saturated monocarboxylic acids that have long carbon chains e.g. **palmitic acid** ($C_{15}H_{31}COOH$) and **stearic acid** ($C_{17}H_{35}COOH$).
- Soap is made by the process of saponification. It is a process that involves conversion of fat or oil into soap and glycerol by the action of heat in the presence of alkali.
- Common soaps are the mixture of salts of higher fatty acids (C_8 to C_{18}) which are manufactured by the saponification of fats.
- Saturated fats give hard soap and vegetative oils (unsaturated fats) give soft soap.

Raw materials used in manufacturing of soap :

(i) Vegetable oils or fats– For manufacturing of soap oils of peanut, mahua, castor oil and coconut are used.

(ii) Castic soda or castic potash– The saponification of oil or fat is performed with the aqueous solution of castic soda (NaOH) or castic potash (KOH).

(iii) Fillers– Fillers are substances that are added to soap to make it more useful for particular applications. For example :

a. Sodium rosinate is added to laundry soap to increase its foaming capacity.

b. Glycerol is added in shaving soaps to prevent them from rapid drying.

c. Sodium silicate increases the durability of the soap and the rapid drying of the soap is prevented and increases the hardners of soap bars.

d. Sulphur is added to produce anti acne soap bars.

Question Bank

1. Which one of the following is used in the manufacture of soaps?

(a) Vegetable oil (b) Mobil oil
(c) Kerosene oil (d) Cutting oil

44th B.P.S.C. (Pre) 2000

Ans. (a)

Soap is a combination of animal fat or plant oil and caustic soda. When dissolved in water, it breaks dirt away from the surface. The modern soap makers use the fat that has been processed into fatty acids. This eliminates many impurities and it produces water as a by-product instead of glycerine. Many vegetable fats including olive oil, palm kernel oil and coconut oil are also used in soap making.

2. Soap, removes grease by –

(a) Coagulation (b) Adsorption
(c) Emulsification (d) Osmosis
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

Emulsifiers are usually long chain compounds with polar groups for example-soap. The soap molecules consist of two parts- (i) long hydrocarbon chain ($C_{17}H_{35}$) which is soluble in fat and (ii) dissolved polar parts in water ($COO^- Na^+$). During emulsion, the dipolar alkaline group of soap dissolves oil or grease and the polar group get dissolved in water. When rubbed, the greases are suspended in the water in the form of small droplets and flows smoothly with water.